

Exploratory Data Analysis using Visualization

Assignment

Dataset: Titanic_Dataset

Problem Statement – Titanic Passenger Analysis

You are a data analyst studying passenger data from the Titanic ship. The dataset contains information about passenger class, age, gender, fare, family members, ticket details, and survival status.

The objective of this assignment is to perform Exploratory Data Analysis (EDA) using Matplotlib and Seaborn to understand passenger distribution, survival patterns, and relationships between different variables using data visualization

You need to create appropriate visualizations, interpret the charts, and generate insights from the data.

Submission Requirements

Students must submit:

1. Google Colab Notebook
2. Visualizations with titles
3. Interpretation for each chart
4. Final insights
5. Clean and readable code

Assignment Tasks

Instructions

Students must:

1. Load the Titanic dataset.
2. Perform basic data exploration using `info()` and `describe()`.
3. Handle missing values if necessary.
4. Create the required visualizations.
5. title and axis labels to each chart.
6. Write interpretation for each visualization.
7. final insights from the dataset.

Assignment Questions (Visualization Based)

Q1. Create a countplot to show the number of passengers by gender.

Answer:

```
sns.countplot(x='Sex', data=df)
plt.title('Number of Passengers by Gender')
plt.xlabel('Gender')
plt.ylabel('Count')
plt.show()
```

✓ **Interpretation:**

- Male passengers are significantly higher than female passengers.

Q2. Create a bar chart to show the number of passengers in each passenger class using different colors.

Answer:

```
sns.countplot(x='Pclass', data=df, palette='Set2')

plt.title('Passenger Count by Class')

plt.xlabel('Passenger Class')

plt.ylabel('Count')

plt.show()
```

✓ **Interpretation:**

- Most passengers belong to **3rd class**, followed by 1st and 2nd.

Q3. Create a histogram to show the distribution of passenger age.

Answer:

```
plt.hist(df['Age'], bins=20)
plt.title('Age Distribution of Passengers')
plt.xlabel('Age')
plt.ylabel('Frequency')
plt.show()
```

✓ Interpretation:

- Most passengers are between 20–40 years.
- Few elderly passengers.

Q4. Create a boxplot to detect outliers in the fare column.

Answer:

```
sns.boxplot(y='Fare', data=df)
plt.title('Fare Distribution with Outliers')
plt.ylabel('Fare')
plt.show()
```

✓ Interpretation:

- Significant outliers exist (very high fares).
- Majority paid lower fares.

Q5. Create a scatterplot to a. show the relationship between age and fare. b. analyze the relationship between family size (SibSp + Parch) and fare.

Answer:

Q5a. Scatterplot: Age vs Fare

```
sns.scatterplot(x='Age', y='Fare', data=df)
plt.title('Age vs Fare')
plt.xlabel('Age')
```

```
plt.ylabel('Fare')
plt.show()
```

✓ **Interpretation:**

- No strong linear relationship between age and fare.
- High fares are spread across age groups.

Q5b. Family Size vs Fare

```
df['FamilySize'] = df['SibSp'] + df['Parch']
```

```
sns.scatterplot(x='FamilySize', y='Fare', data=df)
```

```
plt.title('Family Size vs Fare')
```

```
plt.xlabel('Family Size')
```

```
plt.ylabel('Fare')
```

```
plt.show()
```

✓ **Interpretation:**

- Small families tend to pay moderate fares.
- Very large families are rare.

Q6. Create a countplot to show survival count by gender.

Answer:

```
sns.countplot(x='Sex', hue='Survived', data=df)
```

```
plt.title('Survival Count by Gender')
```

```
plt.xlabel('Gender')
```

```
plt.ylabel('Count')
```

```
plt.show()
```

✓ **Interpretation:**

- Females had a much higher survival rate than males.

Q7. Create a bar chart to show survival by passenger class.

Answer:

```
sns.barplot(x='Pclass', y='Survived', data=df)
plt.title('Survival Rate by Passenger Class')
plt.xlabel('Passenger Class')
plt.ylabel('Survival Rate')
plt.show()
```

✓ **Interpretation:**

- 1st class passengers had the highest survival rate.
- 3rd class had the lowest survival.

Q8. Create a heatmap to show correlation between numerical variables.

Answer:

```
plt.figure(figsize=(10,6))
sns.heatmap(df.corr(numeric_only=True), annot=True, cmap='coolwarm')
plt.title('Correlation Heatmap')
plt.show()
```

✓ **Interpretation:**

- Fare and Pclass are negatively correlated.
- Survival is slightly correlated with Fare and Pclass.

Q9. Identify which passenger class paid the highest fare on average.

Answer:

```
df.groupby('Pclass')['Fare'].mean()
```

1st Class passengers paid the highest average fare.

Q10. Write at least 5 insights from the dataset based on your visualizations.

Answer:

Final Insights

✓ Key Insights:

1. Female passengers had a significantly higher survival rate than males.
2. 1st class passengers were more likely to survive than 2nd and 3rd class.
3. Most passengers were young adults (20–40 years).
4. Fare varies greatly, with many outliers (luxury tickets).
5. Passenger class strongly influenced both fare and survival.
6. The majority of passengers traveled in 3rd class, indicating economic diversity.
7. Family size has limited impact on fare but shows clustering around small groups.